

CMPT 354 Fall 2023

Database Systems I

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Assignment 1

Maximum marks: 100 (5% of the class marks)

Due date: September 24, 2023

Notes

- 1) You must use the exact notation for the RA operators introduced in class. In particular, you cannot use dot notation to refer to attributes but must rename attributes, where necessary.
- 2) If we have evidence of academic dishonesty, in particular use of ChatGPT, in any of the six queries, you will get 0 marks for the entire assignment.

Consider the following schema of a Company database:

```
Employees(eid: integer, ename: string, address: string,  
          supereid: integer, salary: integer)  
Departments(did: integer, dname: string)  
Projects(pid: integer, pname: string, did: integer, budget:  
         integer)  
Works_on(eid: integer, pid: integer, hours: integer)
```

Primary keys are underlined. Each Employee has a supervisor (another Employee) referenced by their supereid. Projects are uniquely assigned to a Department. The Works_on relation records which Employee works on which Project for how many hours a week. The primary key of a relation consists of all its underlined attributes. There are foreign key constraints between Employees.supereid and Employees.eid, between Works_on.eid and Employees.eid, between Works_on.pid and Projects.pid and between Projects.did and Departments.did.

Formulate each of the following queries in relational algebra (RA) or explain why the query cannot be formulated in RA.

Assignment 1.1 (15 marks)

For the “Sales” Department and the “IT” Department, return the eid and the ename of all Employees who work for at least one Project of that Department.

Assignment 1.2 (15 marks)

For each Employee, return their name and the name of their supervisees (the Employees supervised by that Employee).

Assignment 1.3 (15 marks)

Return the eids of Employees who work on a project of every Department, i.e. return the eids of Employees who work for at least one project of every Department.

Assignment 1.4 (15 marks)

Return the eid of the Employees who are working for the maximum number of Projects.

Assignment 1.5 (20 marks)

Return the did of the Departments with a Project that has the maximum budget.

Assignment 1.6 (20 marks)

Return the pids of the Projects for which at least two different Employees are working.

Submission

Submit your solution in pdf-format as assignment1.pdf in CourSys.